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State E-Waste Management Dashboard Report – Rajasthan

Executive Summary

The State E-Waste Management Dashboard is a centralized monitoring system developed to evaluate the generation, collection, transportation, recycling, compliance, environmental impact, awareness initiatives, and overall effectiveness of e-waste management activities across Rajasthan. The dashboard consolidates information from multiple worksheets into a single analytical interface, enabling policymakers and administrators to identify strengths, monitor district-wise performance, and take informed decisions. Based on the dashboard, Rajasthan achieved an overall score of 85.6 out of 100, placing its performance in the 'Excellent' category.

1. Introduction

Electronic waste (e-waste) consists of discarded electrical and electronic equipment such as computers, televisions, mobile phones, batteries, refrigerators, and other appliances. Rapid technological advancements, increasing consumer demand, and shorter product life cycles have significantly increased e-waste generation. Improper disposal leads to environmental pollution because e-waste contains hazardous substances like lead, mercury, cadmium, chromium, and brominated flame retardants. The Rajasthan dashboard helps monitor these issues by tracking generation, collection, recycling, compliance, environmental indicators, and awareness activities.

2. E-Waste Generation Analysis

The dashboard indicates that approximately 7.54 million kilograms of e-waste are generated across the monitored districts. Jaipur contributes the largest share, followed by Jodhpur, Alwar, Kota, and Udaipur due to higher population density, industrial development, and commercial activities. The dataset categorizes waste into IT equipment, consumer electronics, large equipment, lighting equipment, batteries, and miscellaneous electronic waste. Understanding generation patterns enables authorities to allocate collection centres, recycling infrastructure, and awareness programs more effectively.

3. Collection Performance

The collection module compares generated e-waste with officially collected quantities. Around 165 tonnes have been formally collected through authorized channels. The dashboard also tracks collection centres, Extended Producer Responsibility (EPR) targets, and actual achievements. Although urban districts perform relatively better, rural areas still

require additional collection facilities and improved citizen participation. Increasing accessibility to collection centres and encouraging take-back programs can substantially improve collection efficiency.

4. Recycling Infrastructure

Authorized recycling facilities play a critical role in recovering valuable materials such as copper, aluminium, gold, silver, and plastics while preventing hazardous substances from entering the environment. The dashboard lists approximately 21 authorized recycling units with an average material recovery rate of about 86%. Installed processing capacity appears sufficient for current formal collection levels; however, increasing collection would allow better utilization of these facilities and reduce dependence on informal recycling.

5. Compliance Monitoring

The compliance section evaluates producer responsibility under the E-Waste (Management) Rules, 2022. Approximately 17 organizations are fully compliant while 3 are partially compliant. The dashboard records inspections, approvals, penalties, corrective actions, and authorization status. High compliance demonstrates effective regulatory monitoring and encourages environmentally responsible disposal practices among manufacturers and producers.

6. Informal Sector Assessment

A significant portion of e-waste handling still occurs in the informal sector. The dashboard estimates around 1,890 informal workers, of whom approximately 887 have been registered or trained. Formalizing these workers through skill development, certification, and safety training can improve recycling efficiency while reducing occupational health risks caused by exposure to hazardous substances.

7. Environmental and Health Monitoring

Environmental monitoring includes regular assessment of air, soil, and water quality around recycling locations. Health monitoring activities such as medical camps and screening programs have benefited more than 2,048 individuals. Continuous monitoring helps identify pollution hotspots, evaluate the impact of recycling operations, and protect workers as well as nearby communities.

8. Public Awareness Initiatives

The dashboard highlights awareness campaigns including school education programs, community outreach, collection drives, workshops, and industry awareness sessions. Public awareness is essential because proper segregation and disposal begin at the consumer level.

These initiatives encourage citizens to use authorized collection centres instead of disposing of electronic waste with municipal solid waste.

9. Overall Dashboard Assessment

The overall dashboard score of 85.6/100 indicates that Rajasthan has established an efficient e-waste management ecosystem. Strong compliance levels, efficient recycling infrastructure, environmental monitoring, and awareness campaigns contribute to this performance. Nevertheless, improving collection efficiency and integrating more informal workers remain key priorities.

10. Recommendations

Increase the number of authorized collection centres in underserved districts; strengthen implementation of Extended Producer Responsibility; expand awareness campaigns in schools, colleges, industries, and residential societies; integrate informal workers into the formal recycling system; deploy real-time digital monitoring and GIS-based dashboards; promote incentive-based return schemes; and conduct regular environmental audits and health assessments.

11. Conclusion

The Rajasthan State E-Waste Dashboard demonstrates a comprehensive and data-driven approach to sustainable e-waste management. The state's excellent overall score reflects effective governance, strong compliance, and efficient recycling capabilities. Continued investment in collection infrastructure, public participation, and informal sector integration will further strengthen Rajasthan's ability to manage increasing electronic waste in an environmentally responsible manner.